

VFD Threat Categories — Framework Mapping

Threat categories are anchored to three references:

- **MITRE ATT&CK for ICS** — tactic and technique identifiers
- **BSI TR-03183-1** — specific requirement identifiers (REQ_ER)
- **EU Cyber Resilience Act (CRA)** — Annex I essential requirements

Mapping table

#	Threat category	MITRE ATT&CK ICS tactic / technique	BSI TR-03183-1	EU CRA Annex I
1	Unauthorized network access	Initial Access	REQ_ER 5 (Access control), REQ_ER 11 (Limit attack surfaces)	Art. 13 — unauthorized access protection
2	Firmware/software tampering	Persistence — T0839 Module Firmware	REQ_ER 4 (Security updates), REQ_ER 7 (Integrity protection)	§2 — integrity, security updates
3	Denial of service	Inhibit Response Function — T0814	REQ_ER 9 (Availability), REQ_ER 10 (Minimising negative impact)	§6 — availability of essential functions
4	Physical interface exploitation	Initial Access — T0848 Rogue Master	REQ_ER 11.2 (Disable debug interfaces by default)	§8 — limit attack surfaces
5	Command/protocol injection	Impair Process Control — T0836 Modify Parameter, T0855 Unauthorized Command Message	REQ_ER 7 (Integrity), REQ_ER 12 (Incident mitigation)	§3 — integrity of commands and configuration
6	Safety system bypass (STO)	Inhibit Response Function — T0838 Alarm Suppression	REQ_ER 9.3 (Local safety functions during network loss)	§6 — availability and safety functions

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7	Replay attack	Impair Process Control — T0855 Unauthorized Command Message	REQ_ER 6 (Confidentiality), REQ_ER 7 (Integrity)	§3 — integrity
8	Credential/authentication bypass	Initial Access, Lateral Movement	REQ_ER 5 (Access control)	§1 — unauthorized access

Notes

- MITRE ATT&CK ICS tactics referenced: **Initial Access, Persistence, Inhibit Response Function, Impair Process Control, Lateral Movement**
- BSI TR-03183-1 requirements referenced: REQ_ER 4, 5, 6, 7, 9, 9.3, 10, 11, 11.2, 12
- EU CRA Annex I references follow the essential requirements structure (Part 1)
- Threat categories 1 and 8 both map to Initial Access in MITRE, but differ in vector: category 1 is network-based, category 8 is credential-based
- Categories 3 and 6 both map to Inhibit Response Function; category 6 is more specific — it targets the STO hardware safety line directly